

Global Component Design Principle

Components may be coming to a browser near you.

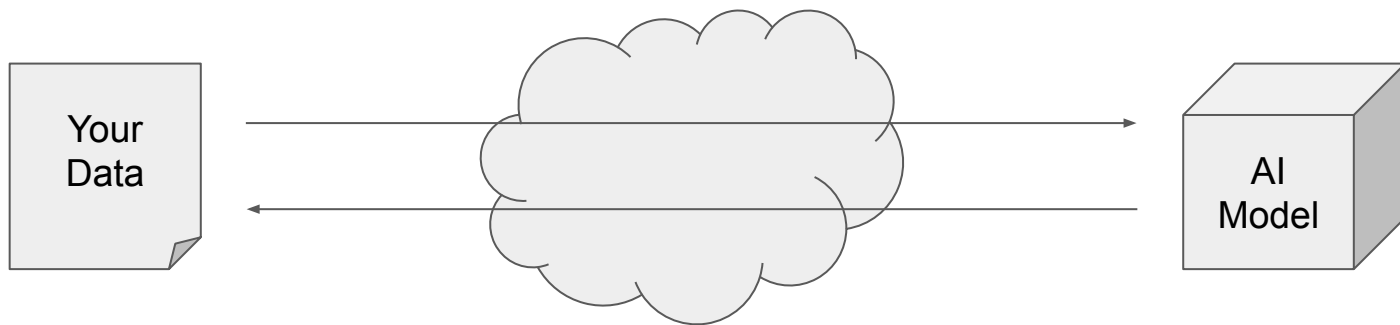
Problem space

On-device AI APIs have been proposed.

Either your data has to come to the model.

Or, the model has to come to you.

(Or both.)



Problem space (by Ehsan)

- Downloaded post-installation of the UA
- Shared across multiple web APIs or features
- Large binary/model artifacts (e.g., ML weights, codecs, language packs)
- Versioned independently of the browser release cycle
- Potentially hardware-accelerated

Examples: Prompt API, Web Speech API, Translator & Language Detector API, ...

Idea: Formulate a design principle for this type of API.

Example: Phi-4-Mini in Microsoft Edge (Prompt API)

```
christianliebel@Christians-MacBook-Pro 2025.10.23.1 % pwd
/Users/christianliebel/Library/Application Support/Microsoft Edge/EdgeLLMOnDeviceModel/2025.10.23.1
christianliebel@Christians-MacBook-Pro 2025.10.23.1 % ls -l
total 4909168
-rw-----@ 1 christianliebel  staff          0 Mar  2 16:37 adapter_cache.bin
-rw-----@ 1 christianliebel  staff        261 Mar  2 14:28 added_tokens.json
-rw-----@ 1 christianliebel  staff        423 Mar  2 14:28 chat_template.jinja
-rw-----@ 1 christianliebel  staff       8794 Mar  2 14:28 edge_on_device_model_execution_config.pb
-rw-----@ 1 christianliebel  staff          0 Mar  2 16:37 encoder_cache.bin
-rw-----@ 1 christianliebel  staff       1722 Mar  2 14:28 genai_config.json
-rw-----@ 1 christianliebel  staff        503 Mar  2 14:28 manifest.json
-rw-----@ 1 christianliebel  staff    2418348 Mar  2 14:28 merges.txt
-rw-----@ 1 christianliebel  staff   26246312 Mar  2 14:28 model.onnx
-rw-----@ 1 christianliebel  staff 2465347584 Mar  2 14:28 model.onnx.data
-rw-----@ 1 christianliebel  staff        617 Mar  2 14:28 special_tokens_map.json
-rw-----@ 1 christianliebel  staff       2652 Mar  2 14:28 tokenizer_config.json
-rw-----@ 1 christianliebel  staff   15524095 Mar  2 14:28 tokenizer.json
-rw-----@ 1 christianliebel  staff   3910310 Mar  2 14:28 vocab.json
```

Web Speech Sample Code (from [design-reviews#1169](#))

```
const meetConfig = {
  langs: ['en-US'],
  processLocally: true,
  // We specifically need a model capable of handling a meeting environment
  quality: 'conversation'
};

async function setupSpeech() {
  // 1. Check if a high-quality model is already available
  const availability = await SpeechRecognition.available(meetConfig);

  if (availability == 'available') {
    startOnDeviceSpeech();
    return;
  }

  // 2. If not, try to install one
  try {
    await SpeechRecognition.install(meetConfig);
    startOnDeviceSpeech();
  } catch (e) {
    // 3. Fallback: Device hardware is too weak for "conversation" quality,
    // or user denied the download. Fallback to Cloud to ensure transcript usability.
    console.log("High-quality on-device model unavailable. Using Cloud.");
    startCloudSpeech();
  }
}
```

Prompt API Review Comment ([design-reviews#1093](#))

“

We're worried about the risks of fingerprinting and excessive disk usage from letting the site select an arbitrary set of languages.

We tentatively believe that local models don't need to be able to understand or produce languages outside the set that the user understands, and possibly the language the page is written in.

Draft Document (by Ehsan)

<https://docs.google.com/document/d/1eXgTEmyR-M-1y50bCeDdH4yvvm8wZYSRWTrrqz0dM6Q/edit?tab=t.0#heading=h.r0fr2p1bcso0>

Thoughts? Opinions?